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FIN302

Finance and Markets

Lecture 8: Futures and Portfolio Hedging

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Lecture 8- Futures and Portfolio Hedging

Questions and Learning Outcomes:

- How to use futures for hedging and risk management?
 - Basis Risk
 - Timing & asset mismatching
 - Cross-Hedging
 - Minimum variance hedging

- How to hedge equity portfolios?
 - The Capital Asset Pricing Model (CAPM)
 - Hedging Market Risk
 - Changing Beta
 - Roll the Hedge Forward



Lecture 8- Futures and Portfolio Hedging

- Readings:
 - Hull (2022)
 - Chapter 3
- Tutorial and Homework Problems from Hull:
 - 3.14, 17, 25, 32, 35

Hedging: Basic Principles

➤ Hedging

- To take a position in futures to **neutralize** or **offset** the risk of the underlying asset as far as possible

➤ Perfect hedge

- Completely eliminate the risk and lock-in buy/sell price

➤ Imperfect hedge

- Basis risk, a common risk in hedging
- What we can do is to perform hedging as close to perfectly

➤ Static hedging: *hedge-and-forget*---no attempt to adjust the hedge once it has been put in place

➤ Dynamic hedging: the hedge is monitored closely and adjusted frequently.

Short Hedge

- A short hedge is appropriate when the hedger already owns the underlying asset and ***expects to sell*** it later
- It involves a **short position in futures contracts**.
- **Example:** On 15 May an oil producer knows he **will sell** one million barrels of crude oil on 15 August.
 - NYMEX crude oil futures are for the delivery of 1,000 barrels.
 - The August contract is trading for \$79 per barrel on 15 May.
 - The producer **short 1,000 contracts on 15 May to hedge his exposure**
 - If S_T is the spot price of crude oil on 15 August and F_T is the August futures price on that date, then $F_T \approx S_T$.
 - On 15 August the producer receives $1,000,000 S_T + 1,000 (79 - F_T)$
 $1,000 \approx \$79,000,000$
 - So he has locked in a price of \$79 per barrel.
 - Comparing to unhedged:
 - this improves when $F_T = S_T < \$79$,
 - but is worse when $F_T = S_T > \$79$.

The Basis

- In a hedging situation, the **basis** is defined by
$$b_t = S_t - F_t$$
 - S_t : spot price of the asset being hedged
 - F_t : futures price of the contract used.
- The basis does not remain constant over time:
 - If $b_1 < b_2$ the basis has strengthened.
 - If $b_1 > b_2$ the basis has weakened

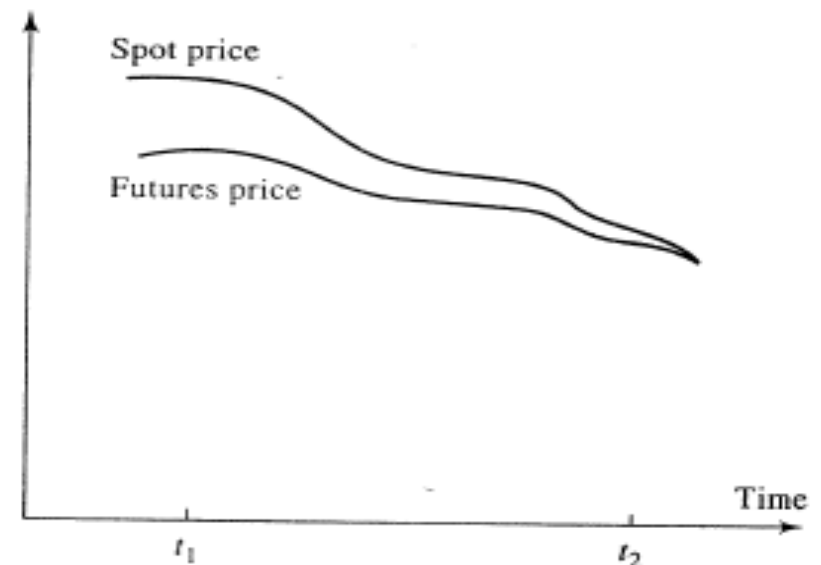


Figure 1: Variation of the basis over time.

Perfect and Imperfect Hedging

- **Perfect hedging:** in the previous two examples, the hedging is perfect because the basis was guaranteed to be zero when the hedge position was closed-out.
- **Imperfect hedging:** in practice, there are two issues
 - **Timing mismatch:** the maturity of futures contract is later than the close-out date of the hedge (WHY?)
 - **Asset mismatch:** the asset to be hedged is different from the underlying asset of the futures
 - e.g., to hedge **jet fuel** price using **heating oil futures**

Basis Risk

Basis Risk:

- In general, it is impossible to know in advance what the basis will be when a hedge is closed-out.
- This uncertainty about the basis is called **basis risk**.
- **Two source of basis risk:**
 - *timing mismatches:*
 - *asset mismatches:*

Basis Risk

- A hedger knows he must **purchase** an asset at time t_2 .
- At time t_1 , he takes a **long** position in a futures contract *maturing after time t_2* , written on another asset
 - S_2^* be the spot price of the asset underlying the futures contract, e.g., **heating oil**
 - S_2 be the spot price of the asset being hedged (at time t_2), e.g., **jet fuel**.
- The effective **purchasing price** of the asset (e.g., jet fuel) at time t_2 :

$$S_2 + (F_1 - F_2) = F_1 + (S_2^* - F_2) + (S_2 - S_2^*)$$

- The effective **buy** price can be decomposed into the initial futures price, plus
 - **timing mismatch:** $(S_2^* - F_2)$
 - **asset mismatch:** $(S_2 - S_2^*)$

Asset Mismatch: Choice of Contract

- The choice of futures contract is vital for managing basis risk.
- This **choice** has two components:
 - ❑ **The choice of underlying asset**
 - The underlying asset of the futures contract matches exactly the asset being hedged---**perfect hedging**.
 - Otherwise, a contract whose price increments are **highly correlated** with the price increments of the asset being hedged,
 - *e.g., to hedge jet fuel price using heating oil futures*
 - imperfect hedging.

Timing Mismatch: Choice of Contract

□ The choice of delivery month

- **The general rule of thumb:** to choose the first contract that matures after the hedge is to be closed-out.
 - It is best ***to avoid futures contracts during their delivery months*** since their prices are often very erratic.
 - The hedger runs the risk of taking delivery of the underlying if he has a long futures position during the delivery month of the contract.
- Sometimes **liquidity considerations** force the hedger to use shorter-dated contracts and ***to roll the hedge forwards***.

Cross-Hedging: Minimum Variance Hedging

- **Cross-hedging:** when the asset underlying the futures contract differs from the asset being hedged.
- **Hedge ratio:** The ratio of the size of the futures position to the size of the exposure being hedged.
 - A hedge ratio of 1.0 is natural *when the asset underlying the futures contract is the same as the asset being hedged.*
 - For cross-hedging, the **minimum-variance hedge ratio**

$$h = \rho \frac{\sigma_S}{\sigma_F}$$

can minimize the variance of his position, where

- σ_S the standard deviation of asset price changes ΔS ,
- σ_F the standard deviation of futures price changes ΔF and
- ρ the correlation between ΔS and ΔF , representing the slope coefficient that arises from regressing ΔS against ΔF

Cross-Hedging: The Optimal Number of Contracts

➤ Note:

$$h = \frac{N_F}{N_A}, \quad N_F = h N_A$$

- Since the portion of the position being hedged is $h Q_A$,
- Q_A the size of the position to be hedged
 - Q_F the size of one futures contract

$$N_F Q_F = h Q_A$$

- Therefore the **number of futures contracts** required is

$$N = N_F = h \frac{Q_A}{Q_F}$$

- N the optimal number of futures contracts for hedging.

Cross-Hedging: Tailing the Hedge

- The expression for the optimal number of contracts does not take into account of the impact of the daily settlement of futures contracts.
- This is corrected by **tailing the hedge**.
 - The adjustment

$$N = h \frac{V_A}{V_F}$$

- $V_A = Q_A S$ the dollar value of the position being hedged
- $V_F = Q_F F$ the dollar value of one futures contract

Cross-Hedging: Example

An airline **will purchase** $Q_A = 2,000,000$ gallons of **jet fuel** after one month, and hedges against a price increase by taking a long position in **heating oil futures**. The spot price of jet fuel and the futures prices of heating oil are $S = \$1.94$ and $F = \$1.99$ per gallon, respectively, when the hedge is initiated.

- Using data one can compute:

$$\sigma_F = 0.0313, \quad \sigma_S = 0.0263, \quad \rho = 0.928$$

- The estimated minimum variance hedge ratio

$$h = \rho \frac{\sigma_S}{\sigma_F} = 0.928 \frac{0.0263}{0.0313} = 0.7777$$

Cross-Hedging: Example

- Each heating oil futures contract is for the delivery of $Q_F = 42,000$ gallons.

- The airline should take a long position in

$$N = h \frac{Q_A}{Q_F} = 0.7777 \frac{2,000,000}{42,000} = 37.03 \approx 37$$

- By tailing the hedge, the airline should take a long position in

$$N = h \frac{V_A}{V_F} = 0.7777 \frac{2,000,000 \times 1.94}{42,000 \times 1.99} = 36.10 \approx 36$$

Hedging Equity Portfolio: CAPM

➤ **Connecting portfolio to the market---**Capital Asset Pricing Model (CAPM)

$$E(R_P) = R_F + \beta_P[E(R_M) - R_F]$$

- R_P : the (total) return on an equity portfolio
- R_M : the (total) return on the market portfolio
- R_F : the risk-free return
- β_P : the beta of the portfolio is given by

$$\beta_P = \frac{Cov(R_P, R_M)}{Var(R_M)} = \rho \frac{\sigma_P}{\sigma_M}$$

measures the portfolio's exposure to systematic risk.

- We can hedge the systematic risk of the portfolio with a position in futures contracts on the index, *a proxy for the market*
- **Note:** Only the systematic risk can be hedged in this way—unsystematic risk can only be eliminated by diversification.



Hedging Equity Portfolio: Hedging Market Risk

- The minimum hedge ratio:

$$h = \rho \frac{\sigma_P}{\sigma_M} = \beta_P$$

- Hedging the systematic risk of the portfolio requires a ***short position in index futures contracts***,

$$N = \beta_P \frac{P}{F}$$

- P the current value of the equity portfolio
- F the current value of one index futures contract

Hedging Equity Portfolio: Example

- Consider a portfolio worth \$5,050,000, with $\beta = 1.5$ relative to the S&P 500 index.
 - The S&P 500 index value is currently 1,000.
 - The three-month S&P 500 index futures are currently trading for 1,010.
 - Each futures contract is \$250 times the index value.
 - The risk-free interest rate is 4% and the dividend yield of the S&P 500 index is 1% per annum.

- Notations:

$$P = 5,050,000, S_0 = 1,000, F_0 = 1,010, R_f = 4\%, F = 250 F_0$$

$$q_I = 1\%, T = \frac{3}{12}$$



Hedging Equity Portfolio: Index Futures

➤ As a fund manager, you want to hedge the systematic risk of the portfolio over the following three months.

- To hedge the systematic risk of the portfolio the manager needs to short

$$N = \beta_P \frac{P}{F} = 1.5 \frac{5,05,000}{1,010 \times 250} = 30$$

- After three months, the value of the S&P 500 index is 900, and the index futures price is 902; i.e. $S_1 = 900, F_1 = 902$
- The gain from the short futures position is
 $V_F = N(F_0 - F_1) = 30 \times (1010 - 902) \times 250 = \$810,000$

Hedging a Portfolio: Example

- The return on the index, ***including the dividend yield***,

$$R_M = \frac{S_1 - S_0}{S_0} + q_I = \frac{900 - 1000}{1000} + 0.0025 = -9.75\%$$

- The expected return on the portfolio is thus

$$\begin{aligned} E(R_P) &= R_F + \beta_P [E(R_M) - R_F] \\ &= 0.01 + 1.5 \times (-0.00975 - 0.01) = -15.125\% \end{aligned}$$

- The expected value of the portfolio, ***inclusive of dividends***, after three months is therefore

$$V_P = P[1 + E(R_p)] = 5,050,000 \times (1 - 0.15125) = \$4,286,187$$

- So the expected value of the fund manager's position, including the gain from the futures contracts, is

$$V = V_F + V_P = 4,286,187 + 810,000 = \$5,096,187$$

- Once a portfolio has been hedged with index futures ***the total beta becomes zero***.

Hedging a Portfolio: Example

- Table 1 presents the outcomes for several other scenarios.

Value of index in three months:	900	950	1,000	1,050	1,100
Futures price of index today:	1,010	1,010	1,010	1,010	1,010
Futures price of index in three months:	902	952	1,003	1,053	1,103
Gain on futures position (\$):	810,000	435,000	52,500	-322,500	-697,500
Return on market:	-9.750%	-4.750%	0.250%	5.250%	10.250%
Expected return on portfolio:	-15.125%	-7.625%	-0.125%	7.375%	14.875%
Expected portfolio value in three months including dividends (\$):	4,286,187	4,664,937	5,043,687	5,422,437	5,801,187
Total value of position in three months (\$):	5,096,187	5,099,937	5,096,187	5,099,937	5,103,687

Hedging a Portfolio: Changing Beta

- Instead of hedging all the systematic risk of the portfolio, it may be desirable simply to adjust its beta β to some target β^* .
- This is achieved by taking a position in

$$N^* = (\beta - \beta^*) \frac{P}{F}$$

- If $\beta > \beta^*$ then $N^* > 0$, which corresponds to a **short** futures position, **reducing the market risk** of the portfolio, keeping a limited exposure to the market.
- If $\beta < \beta^*$ then $N^* < 0$, which corresponds to a **long** futures position, **increasing the market risk** of the portfolio, hoping to benefit from bull market (for speculation, instead of hedging).

Hedging a Portfolio: Changing Beta

Example: Consider the previous example.

- Suppose you decide to reduce the beta of this portfolio to $\beta^* = 0.75$. You then take a **short** position for 15 index futures

$$N^* = (\beta - \beta^*) \frac{P}{F} = (1.5 - 0.75) \frac{5,050,000}{252,500} = 15$$

- Alternatively, if you decide to increase the beta to $\beta^* = 2$, then you should take a **long** position for 10 index futures

$$N^* = (\beta - \beta^*) \frac{P}{F} = (1.5 - 2) \frac{5,050,000}{252,500} = -10$$

Stack and Roll

- Sometimes the close-out date for a hedge exceeds the maturities of all available futures contracts.
 - In this case the hedger must **roll the hedge forward** by repeatedly closing out one contract and taking the same position in a contract with a later maturity.
 - ***Rolling a hedge***, therefore, generates random profits/losses on the roll-over dates, which represent a type of ***basis risk***.
- In any hedging situation there is a danger that ***losses will be realized on the hedge*** while the ***gains on the underlying exposure are unrealized***
- This can create ***liquidity problems***

Stack and Roll

Example: In **April 2021** a company realizes it will have 100,000 barrels of oil to sell in **June 2022**. It hedges its risk with oil futures, using a hedge ratio of 1.0 (i.e. no tailing is made).

- Although oil futures exist for delivery each month, only the contracts maturing in the next six months are sufficiently liquid.
- Each contract is for the delivery of 1,000 barrels.
- The company begins by shorting 100 **October 2021** contracts in April 2021.
- In **September 2021** it rolls the hedge forward into the **March 2022** contract.
- In **February 2022** it rolls the hedge forward again into the **July 2022** contract.

Stack and Roll

- One possible outcome is shown in Table 3.5. There we see that the company gains
 $(48.20 - 47.40) + (47.00 - 46.50) + (46.30 - 45.90) = \1.70
per barrel from its futures strategy.
- However, during the period from April 2021 to June 2022 the spot price for oil declines from \$49 to \$46 per barrel.
- So the company receives \$1.70 per barrel in compensation for a \$3 per barrel decline in the oil price.

Table 3.5 Data for the example on rolling oil hedge forward.

<i>Date</i>	<i>Apr. 2021</i>	<i>Sept. 2021</i>	<i>Feb. 2022</i>	<i>June 2022</i>
Oct. 2021 futures price	48.20	47.40		
Mar. 2022 futures price		47.00	46.50	
July 2022 futures price			46.30	45.90
Spot price	49.00			46.00

Appendix: Additional Reading

Hedging a Portfolio: Market Risk

- *Hedging a portfolio with equity index futures is generally better than selling it and investing in risk-free instruments:*
 - Hedging with futures *removes systematic risk*.
 - It allows for generating *a positive return* (around a risk-free rate), irrespective of the change in the market.
 - Most equity portfolios are held with a very *long-term* view.
 - Liquidating a large equity portfolio and investing the proceeds in risk-free securities would almost certainly *breach investor mandates and investment policies*.

Cross-Hedging: Minimum Variance Hedging

➤ Minimum-variance hedge ratio

$$h = \frac{N_F}{N_A} = \rho \frac{\sigma_S}{\sigma_F}$$

Short hedge: sell N_A asset at t_2 and short N_F futures at t_1 .

- The value of the hedging portfolio at t_2 :

$$Y = S_2 N_A + (F_1 - F_2) N_F = S_1 N_A + (S_2 - S_1) N_A - (F_2 - F_1) N_F$$

i.e.,

$$Y = S_1 N_A + N_A \left(\Delta S - \Delta F \frac{N_F}{N_A} \right) = S_1 N_A + N_A (\Delta S - h \times \Delta F)$$

- The variance of Y :

$$\begin{aligned} \text{Var}(Y) &= (N_A)^2 [\sigma_S^2 + h^2 \sigma_F^2 - 2h \rho \sigma_S \sigma_F] \\ &= (N_A)^2 [(h\sigma_F - \rho\sigma_S)^2 + \sigma_S^2 - \rho\sigma_S^2] \end{aligned}$$

- Therefore, the minimum variance is achieved when

$$h\sigma_F = \rho\sigma_S$$

